Spectral reference	Sequence name	Sequence	Reference	Origin
Parallel DNA	1R3X	AGAGAGAA + TTCTCTCT + TTCTCTCT	10.1021/ja973221m	Synthetic
	Parallel DNA	GAAGAAGAAGAAG + CTTCTTCTTCTTC	10.1002/j.1460-2075.1992.tb05463.x	Synthetic
	Triplex Py:Pu-Pu intra 1	TTCTTCTTCTTCTTCTTTTGAAGAAGAAGAAGAATTTTAAGAAGAAGAAGAAG	10.1093/nar/gkh274	Synthetic
	Triplex Py:Pu-Pu intra 2	TTCTTCTTCTTCTTCTTTTGAAGAAGAAGAAGAATTTTAAGAAGAAGAAGAAG	10.1093/nar/gkh274	Synthetic
	Triplex Py:Pu-Py intra 2	CTTCTTCTTCTTCTTTTTTTTCTTCTTCTTCTTCTTTTGAAGAAGAAGAAGAA	https://www.ncbi.nlm.nih.gov/books/NBK6545/	Synthetic
Double helix DNA (B-form)	1CS2	CTAAAGCAGTAG + CTACTGCTTTAG	10.1093/nar/27.24.4759	Synthetic
	Br1	TCGTAGCAAGCT + AGCTTGCTACGA		Synthetic
	Br2	GCTCTAGCACCG + CGGTGCTAGACG		Synthetic
	Br3	CCGCTTAAACGC + GCGTTTAAGCGG		Synthetic
	DsrA 22 ADN 10mM	AAGTGCTTCTTGCTTAAGCAAG	10.18388/abp.2014_1941	Synthetic
	B-form-60	E. coli Genomic DNA	10.1002/bip.1979.360180418	Literature
Double helix RNA (A-form)	1RNK	GGCGCAGUGGGCUAGCGCCACUCAAAAGGCCCAU	10.2210/pdb1RNK/pdb	Synthetic
	DsrA 14 ARN	CUUGCUUAAGCAAG	10.18388/abp.2014_1941	Synthetic
	DsrA 22 ARN	AAGUGCUUCUUGCUUAAGCAAG		Synthetic
	Pseudoknot	GGGCUGUUUUUCUCGCUGACUUUCAGCCCCAAACAAAAAAGUCAGCA	10.1016/j.molcel.2005.01.017	Synthetic
	rpsO	CAGAAAAGGGGGCCUGAGUGGCCCCUUUUUUCAA	10.1111/febs.70445	Synthetic
	tRNA	GCGGAUUUAGCUCAGUUGGGAGAGCGCCAGACUGAAGAUCUGGAGGUCCUGUGUUCGAUCCACAGAAUUCGCACCA	10.1038/s41467-019-11356-w	Synthetic
DNA Parallel quadruplex	(G3T)4	GGGTGGGTGGGTGGGT		Synthetic
	2M27	CGGGGCGGGCCTTGGGCGGGGT	10.1093/nar/gkt784	Synthetic
	TT(GGGT)4	TTGGGTGGGTGGGTGGGT		Synthetic
	cMycparallel	TGAGGGTGGGTAGGGTGGGTAA	10.1021/bi048242p	Synthetic
DNA Hybrid quadruplex	2F8U	GGGCGCGGGAGGAATTGGGCGGG	10.1093/nar/gkl610	Synthetic
	2HY9	AAAGGGTTAGGGTTAGGGTTAGGGAA	10.1093/nar/gkm009	Synthetic
	2LOD	GGGATGGGACACAGGGGACGGG	10.1093/nar/gks329	Synthetic
Z-DNA	d(CG)9	CGCGCGCGCGCGCGCGCG	10.1016/j.bmcl.2023.129376	Synthetic
	poly(GC)	(GC)n		Literature
	Z-form	(GC)n	10.1093/nar/13.13.4983	Literature
RNA quadruplex	GGA4	GGAGGAGGAGGA	10.1080/15476286.2025.2512613	Synthetic
	GGU4	GGUGGUGGUGGU	10.1080/15476286.2025.2512613	Synthetic
	RNAquad	UGGCGGGAGGAGGC	10.1093/nar/gkae927	Synthetic
DNA Cruciform prone	1DCW	CCGGTACCGG	10.1073/pnas.97.8.3971	Synthetic
	467D	CCGGGACCGG	10.1038/13277	Synthetic
	ctrl5metC	CCTGGCCAGGCCTGGCCAGG	10.1002/anie.202016822	Synthetic
DNA i-motif	iC3-1	CCCACCCAAACCCACCCA		Synthetic
	iC3-2	CCCACCCAAACCCACCCA		Synthetic
	iC5-1	TCCCCC		Synthetic
	iC5-2	CCCCCAACCCCCTTTTTCCCCCTTCCCCC		Synthetic
	iC5-3	CCCCCTTCCCCCAAAAACCCCCCTTCCCCC		Synthetic
	iC5-4	CCCCCAACCCCCAAAAACCCCCAACCCCC		Synthetic
	i-motif ph5	CCCACCCACCCACCCA	10.1038/363561a0	Synthetic
AU rich RNA	r(A5U5)	(AAAAAUUUUUAAAAAUUUUU)		Synthetic
	rA20rU20	AAAAAAAAAAAAAAAAAAAA + UUUUUUUUUUUUUUUUUUUU		Synthetic
	r(A3U3)	(AAAUUUAAAUUUAAAUUUAAAUUUAAAUUU)		Synthetic
AT rich DNA	(A5T5A5T5)n	(AAAAATTTTTAAAAATTTTTAAAAATTTTTAAAAATTTTTAAAAATTTTTAAAAATTTTTAAAAATTTTT)		Synthetic
	A20(AT)20T20	(AAAAAAAAAAAAAAAAAAAAATATATATATATATATATTTTTTTTTTTTTTTTTTTT)		Synthetic
	d(A3T3)	(AAATTTAAATTTAAATTTAAATTTAAATTT)		Synthetic
	d(A2T2)	(AATTAATTAATTAATTAATTAATT)		Synthetic

	Curved B-DNA Sinden	(CAAAATTTTG)	10.1016/S1067-5701(98)80019-3	Synthetic
A-DNA	A-form-59	C. perfringens Genomic DNA	10.1002/bip.1979.360180418	Literature
	A-form-60	E. coli Genomic DNA		Literature
	A-form-61	M. luteus Genomic DNA		Litteratue
Antiparallel quadruplex 1	148D Lithium defolded	GGTTGGTGTGGTTGG	10.1006/jmbi.1994.1105	Synthetic
	148D	GGTTGGTGTGGTTGG		Synthetic
	TBA Quadruplex	GGTTGGTGTGGTTGG	10.1016/j.ijbiomac.2018.09.102	Synthetic
Antiparallel quadruplex 2	2KM3 Lithium defolded	AGGGCTAGGGCTAGGGCTAGGG	10.1093/nar/gkp630	Synthetic
	2KM3	AGGGCTAGGGCTAGGGCTAGGG		Synthetic
	22CTA	AGGGCTAGGGCTAGGGCTAGGG	10.1093/nar/gkp630	Synthetic
Complex oligonucleotide	7CLS	TTGGATCTGAGAATCAGATGTGGGTGGGTGGGT	10.1093/nar/gkaa738	Synthetic
	8DUT	CTATGTATACAAAGAGGGTGGGTAGGGTGGGTTTAATGCGGCACGC + GCGTGCCGCATTAATTTTTTTTTTTTTTTTTTTTTTTTGTATACATAG	10.1093/nar/gkad014	Synthetic
	7O5E	CCCGTTTCCTCGCGAAGCATTCGCGCCCGTTTCCT	10.1021/jacs.1c04679	Synthetic
	8S4N	CGCTCACGCCCCGTTTCCCCCCCCCCCGTTTCCCCCGTGAGCG	10.1093/nar/gkae1282	Synthetic
	6R9K	GCGTGGGTCAGGGTGGGTGGGACGC	10.1002/anie.201905372	Synthetic
	6R9L	GCGTGGTTCAGGGTGGGTGGGACGC		Synthetic
	8ABD	TTAGGTGGGTAGGGTGGGCTAGTCATTTTGACTAGG	10.1093/nar/gkac1088	Synthetic
	7PNE	TAGGTGGGTAGGGTGGGCTAGTCATTTTGACTAGG	10.1002/chem.202103718	Synthetic
	4V2S	UUCCGAUGUAGACCCGUCCUCCUUCGCCUGCGUCACGGGUCCUGGUUAGACGCAGGCGUUUUCUG	10.7554/eLife.05375	Synthetic
	rpsO-polyA	CAGAAAAGGGGGCCUGAGUGGCCCCUUUUUUCAAAAAAAAAAAAAAAAA	10.1111/febs.70445	Synthetic